

Functions are one of the most important ideas in mathematics. They allow us to describe how one quantity depends upon another. Functions appear throughout engineering, science and technology. For example, the distance travelled by a train depends upon its speed and the time for which it travels. Similarly, the voltage across an electrical component may depend upon the current flowing through it. Many of the equations studied in the previous chapters are examples of functions. In this chapter we shall formalise the idea of a function and learn how to work with function notation, tables and graphs.

3.1 What is a Function?

Definition

A function describes how one variable depends on one or more other variables. For every permitted input value there must be exactly one output value.

For example,

$$y = 2x + 3$$

is a function because every value of x produces exactly one value of y . When $x = 1$,

$$y = 2(1) + 3 = 5$$

and when $x = 4$,

$$y = 2(4) + 3 = 11$$

The same rule is simply applied to different values of x . A useful way to think about a function is as a machine:

$$\text{Input} \rightarrow \text{Rule} \rightarrow \text{Output}$$

The input enters the function, the function rule is applied and the output is produced.

3.1.1 Independent and Dependent Variables

Functions describe relationships between variables.

Definition

The independent variable is the variable whose value is chosen. The dependent variable is the variable whose value depends upon the independent variable.

Consider

$$y = 2x + 3$$

The value of x can be chosen freely. The value of y depends entirely upon the chosen value of x . Therefore:

- x is the independent variable.
- y is the dependent variable.

Engineering Application

Suppose a train travels at a constant speed of 20 m/s. The distance travelled is given by

$$d = 20t$$

where d is the distance travelled and t is the time. As the time increases, the distance increases. Here t is the independent variable and d is the dependent variable.

3.1.2 Function Notation

Functions are often written using function notation. Instead of writing

$$y = x^2 + 2x$$

we may write

$$f(x) = x^2 + 2x$$

The notation $f(x)$ means **the value of the function when the input is x** . A common mistake is to think that $f(x)$ means $f \times x$. It does not. The symbols f , g and h are simply names given to functions.

$$f(x), \quad g(x), \quad h(x)$$

are all valid function names.

3.1.3 Evaluating Functions

Evaluating a function means substituting a value into the function and calculating the corresponding output.

Definition

To evaluate a function:

1. Substitute the given value.
2. Use brackets when substituting negative numbers.
3. Simplify using BIDMAS.

Worked Example

Given $f(x) = x^2 + 2x$, find $f(3)$.

Substituting $x = 3$,

$$\begin{aligned}f(3) &= 3^2 + 2(3) \\&= 9 + 6 \\&= 15\end{aligned}$$

Worked Example

Given $f(x) = x^2 + 2x$, find $f(-2)$.

Substituting $x = -2$,

$$\begin{aligned}f(-2) &= (-2)^2 + 2(-2) \\&= 4 - 4 \\&= 0\end{aligned}$$

Worked Example

Given $g(x) = 3x - 5$, find $g(-1)$.

Substituting $x = -1$,

$$\begin{aligned}g(-1) &= 3(-1) - 5 \\&= -3 - 5 \\&= -8\end{aligned}$$

3.1.4 Tables of Values

A function may be represented using a table of values. Consider

$$y = 2x + 1$$

Substituting several values of x gives:

x	y
0	1
1	3
2	5
3	7
4	9

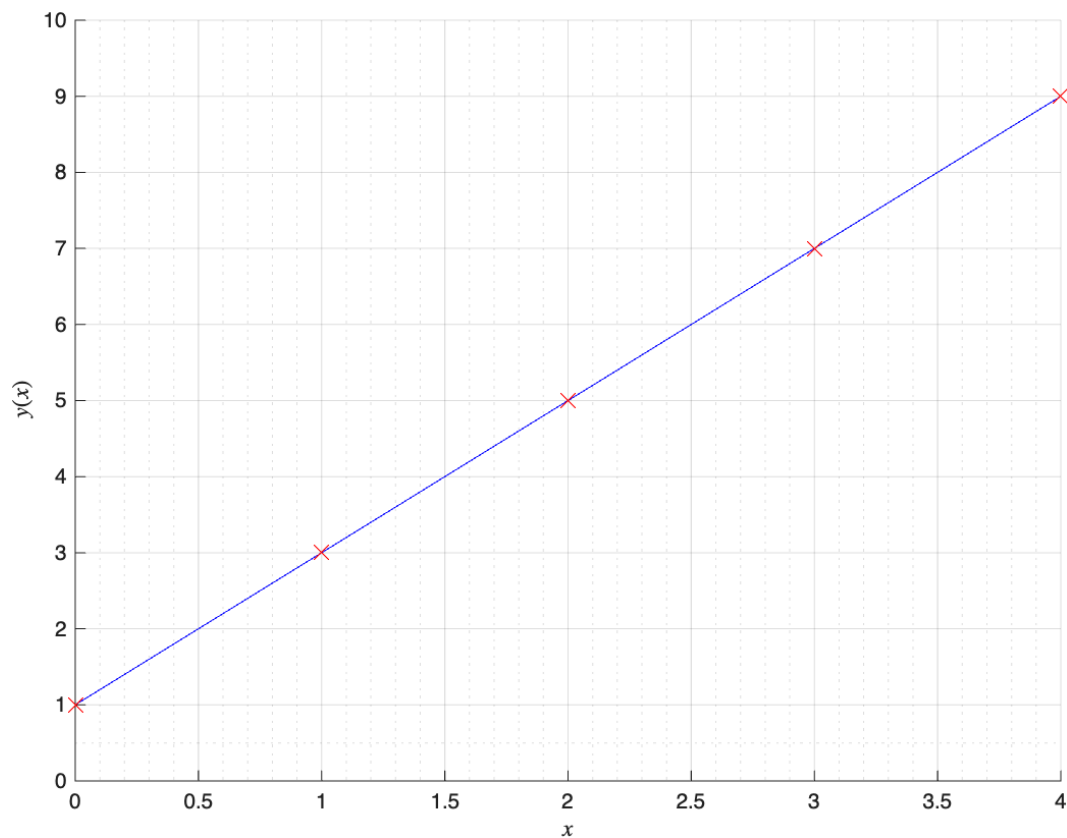
Tables provide a simple way to explore how a function behaves. They are also useful when sketching graphs.

3.1.5 Function Graphs

A graph provides a visual representation of a function. Each point on the graph corresponds to an input-output pair. For example, the table

$$(0, 1), (1, 3), (2, 5), (3, 7)$$

can be plotted on a graph and joined to form a straight line.



Different functions produce different graph shapes. For example:

- Constant functions produce horizontal lines.
- Linear functions produce straight lines.
- Quadratic functions produce curved parabolas.

These functions will be studied in detail later in the chapter.

3.1.6 Domain and Range

Not every function accepts every possible input value. Similarly, not every output value may be possible.

Definition

The domain of a function is the set of allowable input values.

The range of a function is the set of possible output values.

3.1.6.1 Example: Linear Function

Consider

$$f(x) = 2x + 1$$

Any real number can be substituted for x . Therefore the domain consists of all real numbers and the range consists of all real numbers.

3.1.6.2 Example: Quadratic Function

Consider

$$f(x) = x^2$$

Any real number may be substituted. However, squaring a number can never produce a negative value. Therefore:

- Domain: all real numbers.
- Range: $y \geq 0$.

3.1.6.3 Example: Square Root Function

Consider

$$f(x) = \sqrt{x}$$

Negative values cannot be placed under the square root sign. Therefore:

- Domain: $x \geq 0$.
- Range: $y \geq 0$.

3.1.7 Is it a Function?

The defining property of a function is that every input must produce exactly one output.

Worked Example

Consider the following table.

x	y
1	4
2	7
3	10

Each input is associated with exactly one output. Therefore, this represents a function.

Worked Example

Consider the following table.

x	y
2	5
2	8

The input 2 produces two different outputs. Therefore, this does not represent a function.

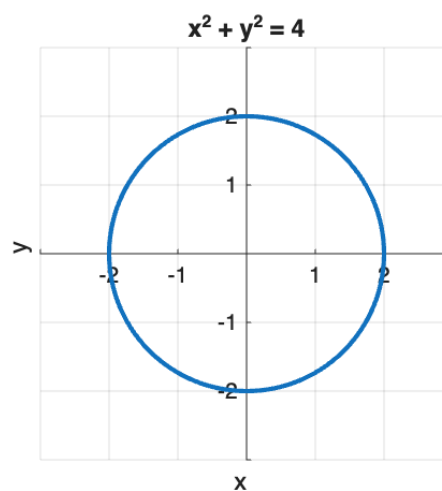
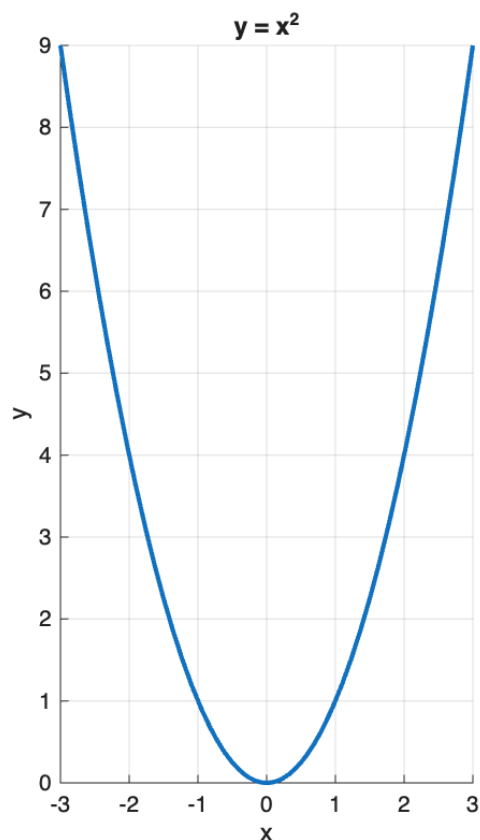
The same idea can be applied to graphs.

Definition

A graph represents a function if every value of x corresponds to only one value of y .

For example:

- A straight line is a function.
- A parabola is a function.
- A circle is not a function, as some values of x correspond to two values of y .



3.1.8 Common Mistakes

Common errors when working with functions include:

- Thinking that $f(x)$ means $f \times x$.
- Forgetting brackets when substituting negative values.
- Confusing the independent and dependent variables.
- Assuming every equation represents a function.
- Forgetting that each input must have only one output.
- Confusing domain and range.

3.1.9 Engineering Application

Engineering Application

Braking distance is often approximated using a quadratic relationship of the form

$$d = kv^2$$

where d is the braking distance, v is the vehicle speed, k is a constant. Suppose

$$d = 0.08v^2$$

If $v = 20$ m/s, then

$$d = 0.08(20)^2$$

$$= 0.08(400)$$

$$= 32 \text{ m}$$

This function shows how braking distance depends upon vehicle speed.

3.1.10 Exercises

Exercise

1. Given

$$f(x) = 2x + 5$$

calculate:

(a) $f(3)$

(b) $f(7)$

(c) $f(-2)$

2. Given

$$g(x) = x^2 - 4x$$

find:

(a) $g(2)$

(b) $g(5)$

(c) $g(-1)$

3. State the independent and dependent variables in

$$C = 3t + 12$$

4. Complete the table of values for

$$y = 2x - 1$$

x	y
0	
1	
2	
3	

5. State the domain and range of

$$y = x^2$$

6. State the domain and range of

$$y = \sqrt{x}$$

Exercise

7. Sketch the graph of $y = 3x - 2$
8. Sketch the graph of $y = x^2$
9. Which of the following represent functions?
 - (a) A straight line.
 - (b) A parabola.
 - (c) A circle.
10. A train travels at a constant speed of 30 m/s. Write a function relating distance d to time t . Hence determine the distance travelled after:
 - (a) 10 s
 - (b) 20 s
 - (c) 60 s

Solutions to Exercise 3.1.10

1. (a) Substitute $x = 3$

$$\begin{aligned}f(3) &= 2(3) + 5 \\&= 6 + 5 \\&= 11\end{aligned}$$

- (b) Substitute $x = 7$

$$\begin{aligned}f(7) &= 2(7) + 5 \\&= 14 + 5 \\&= 19\end{aligned}$$

- (c) Substitute $x = -2$

$$\begin{aligned}f(-2) &= 2(-2) + 5 \\&= -4 + 5 \\&= 1\end{aligned}$$

2. (a) Substitute $x = 2$

$$\begin{aligned}g(2) &= 2^2 - 4(2) \\&= 4 - 8 \\&= -4\end{aligned}$$

- (b) Substitute $x = 5$

$$\begin{aligned}g(5) &= 5^2 - 4(5) \\&= 25 - 20 \\&= 5\end{aligned}$$

- (c) Substitute $x = -1$

$$\begin{aligned}g(-1) &= (-1)^2 - 4(-1) \\&= 1 + 4 \\&= 5\end{aligned}$$

3. In the equation $C = 3t + 12$ the value of C depends upon the value of t . Therefore,

- t is the independent variable.
- C is the dependent variable.

4. Using $y = 2x - 1$

$x = 0$	$y = 2(0) - 1 = -1$
$x = 1$	$y = 2(1) - 1 = 1$
$x = 2$	$y = 2(2) - 1 = 3$
$x = 3$	$y = 2(3) - 1 = 5$

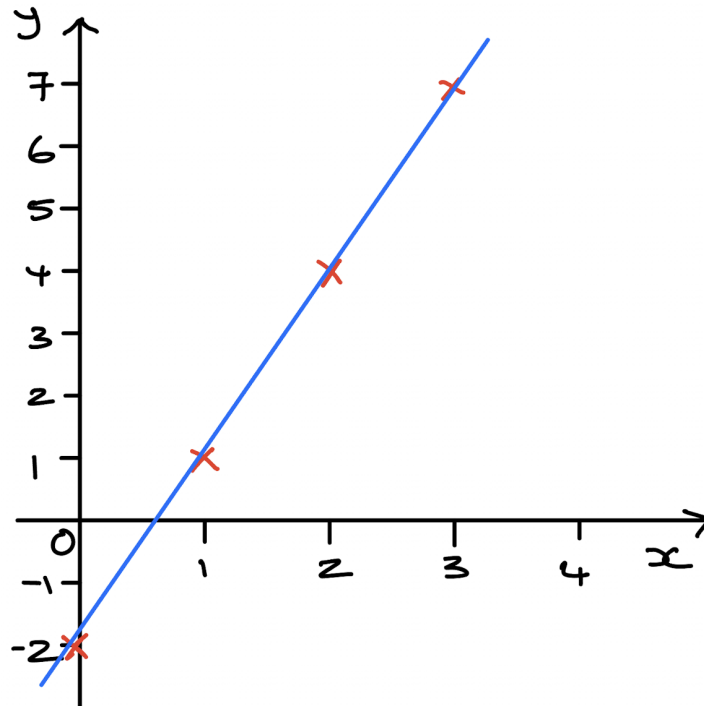
Therefore:

x	y
0	-1
1	1
2	3
3	5

5. For $y = x^2$ any real value may be substituted for x . Therefore the domain is all real numbers. Squaring any real number produces a value greater than or equal to zero. Therefore the range is $y \geq 0$.
6. For $y = \sqrt{x}$ the value inside the square root cannot be negative. Therefore, Domain = $\{x : x \geq 0\}$. The square root of a non-negative number is also non-negative, so Range = $\{y : y \geq 0\}$.
7. To sketch the graph of $y = 3x - 2$ calculate several points:

x	y
0	-2
1	1
2	4
3	7

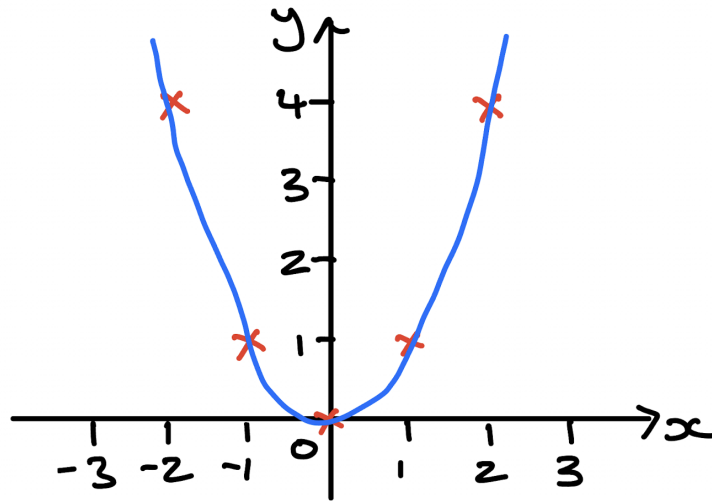
Plot the points $(0, -2)$, $(1, 1)$, $(2, 4)$, $(3, 7)$ and join them with a straight line.



8. To sketch the graph of $y = x^2$ calculate several points.

x	y
-2	4
-1	1
0	0
1	1
2	4

Plot the points $(-2, 4)$, $(-1, 1)$, $(0, 0)$, $(1, 1)$, $(2, 4)$ and join them with a smooth curve. The graph is a parabola opening upwards.



9. (a) Every value of x corresponds to exactly one value of y . Therefore, a straight line represents a function.
- (b) Every value of x corresponds to exactly one value of y . Therefore, a parabola represents a function.
- (c) Some values of x correspond to two values of y . For example, on a circle centred at the origin, $x = 0$ corresponds to both the upper and lower points of the circle. Therefore, a circle does not represent a function.
10. (a) After 10 s,

$$\begin{aligned} d &= 30(10) \\ &= 300 \text{ m} \end{aligned}$$

- (b) After 20 s,

$$\begin{aligned} d &= 30(20) \\ &= 600 \text{ m} \end{aligned}$$

- (c) After 60 s,

$$\begin{aligned} d &= 30(60) \\ &= 1800 \text{ m} \end{aligned}$$